

● HARD

● PROBLEM-SOLVING AND DATA ANALYSIS

Mixed

10 questions · ~15 min

04

Q1

GRID-IN

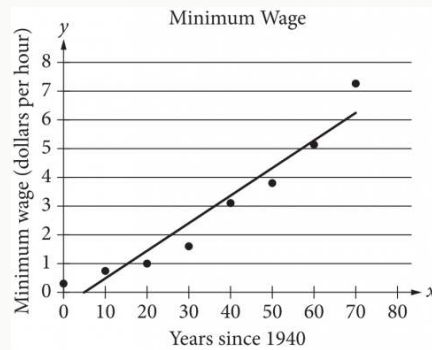
PROBLEM-SOLVING AND DATA ANALYSIS

The number w is 110% greater than the number z . The number z is 55% less than 50. What is the value of w ?

STUDENT-PRODUCED RESPONSE

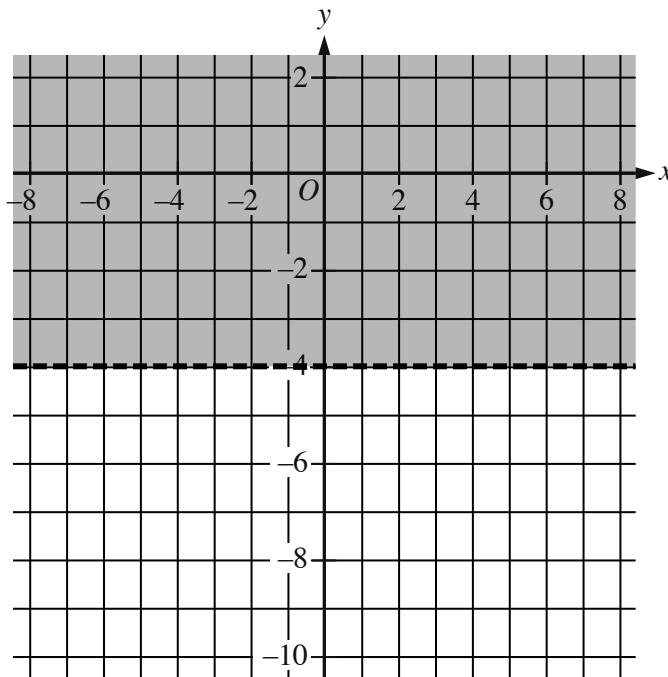
.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



The scatterplot above shows the federal-mandated minimum wage every 10 years between 1940 and 2010. A line of best fit is shown, and its equation is $y = 0.096x - 0.488$. What does the line of best fit predict about the increase in the minimum wage over the 70-year period?

- A. Each year between 1940 and 2010, the average increase in minimum wage was 0.096 dollars.
- B. Each year between 1940 and 2010, the average increase in minimum wage was 0.49 dollars.
- C. Every 10 years between 1940 and 2010, the average increase in minimum wage was 0.096 dollars.
- D. Every 10 years between 1940 and 2010, the average increase in minimum wage was 0.488 dollars.



- The boundary of the inequality is a dashed line.
 - The line is horizontal.
 - The line passes through the following points:
 - (negative 3 comma negative 4)
 - (0 comma negative 4)
 - (3 comma negative 4)
- The area above the boundary is shaded.

The shaded region shown represents the solutions to the inequality $ry < 60$, where r is a constant. What is the value of r ?

- A. 15
- B. 4
- C. -4

D. -15

Q4

ALGEBRA

$$-12x + 14y = 36$$

$$-6x + 7y = -18$$

How many solutions does the given system of equations have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

x	fx
-4	0
$-\frac{19}{5}$	1
$-\frac{18}{5}$	2

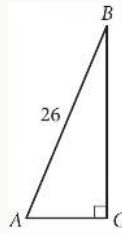
For the linear function f , the table shows three values of x and their corresponding values of fx . If $hx = fx - 13$, which equation defines h ?

- A. $hx = 5x - 4$
- B. $hx = 5x + 7$
- C. $hx = 5x + 9$
- D. $hx = 5x + 20$

Q6

GRID-IN

GEOMETRY AND TRIGONOMETRY



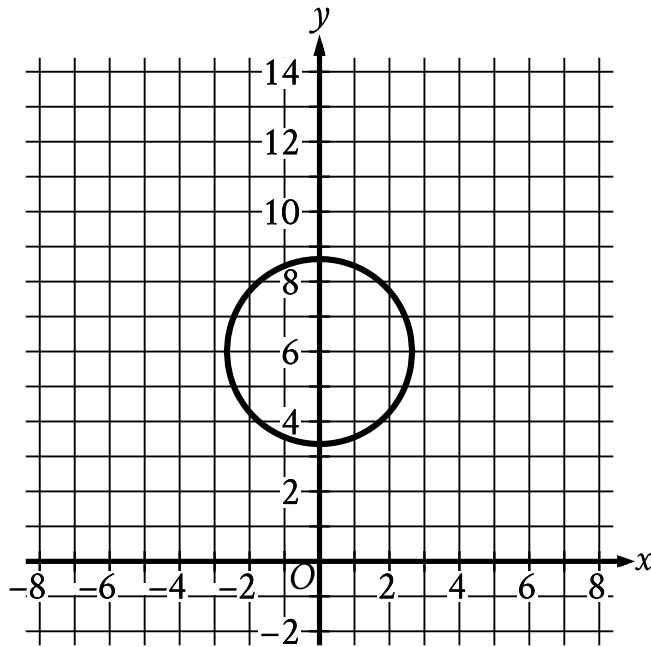
Triangle ABC above is a right triangle, and $\sin(B) = \frac{5}{13}$. What is the length of side $\overline{BC}$?

STUDENT-PRODUCED RESPONSE

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☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
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☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- The circle passes through the following approximate points:
 - (0 comma 8.6)
 - (2.6 comma 6)
 - (0 comma 3.4)
 - (negative 2.6 comma 6)

Circle A shown is defined by the equation $x^2 + y - 6^2 = 7$. Circle B (not shown) has the same radius but is translated 96 units to the right. If the equation of circle B is $x - h^2 + y - k^2 = a$, where h , k , and a are constants, what is the value of $4a$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
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1	1	1	1
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3	3	3	3

4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$D = T - \frac{9}{25}(100 - H)$$

The formula above can be used to approximate the dew point D , in degrees Fahrenheit, given the temperature T , in degrees Fahrenheit, and the relative humidity of H percent, where $H > 50$. Which of the following expresses the relative humidity in terms of the temperature and the dew point?

A. $H = \frac{25}{9}(D - T) + 100$

B. $H = \frac{25}{9}(D - T) - 100$

C. $H = \frac{25}{9}(D + T) + 100$

D. $H = \frac{25}{9}(D + T) - 100$

$$57x^2 + (57b + a)x + ab = 0$$

In the given equation, a and b are positive constants. The product of the solutions to the given equation is kab , where k is a constant. What is the value of k ?

A. $\frac{1}{57}$

B. $\frac{1}{19}$

C. 1

D. 57

In the xy -plane, the graph of $y = 3x^2 - 14x$ intersects the graph of $y = x$ at the points $(0, 0)$ and (a, a) . What is the value of a ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
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3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.